

LAB - Consultas en SQL

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Consulta de un tabla completa

public <postgres 2> Script-10 X

select * from "Materiales" m

Materiales 1 X

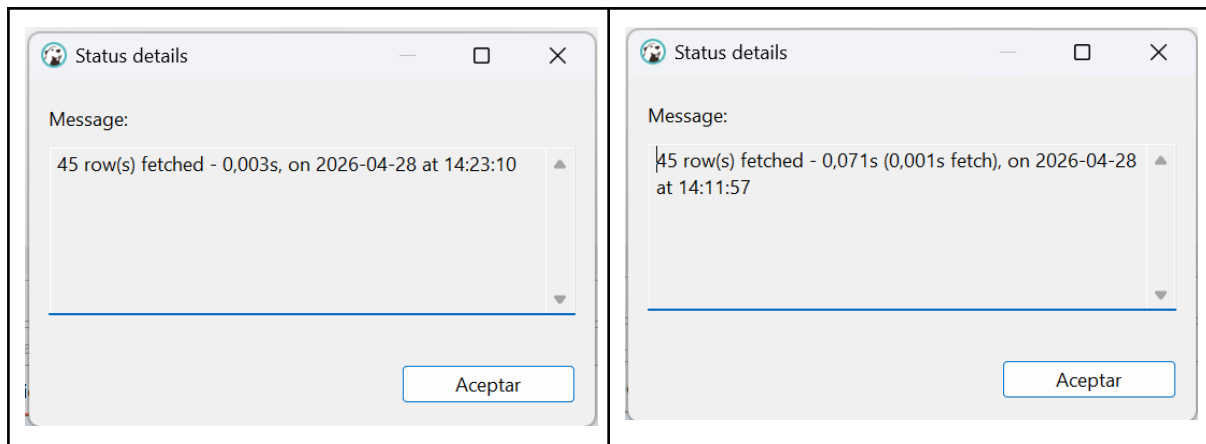
select * from Materiales Enter a SQL expression to filter results (use Ctrl+Space)

Grilla	123 clave	AZ descripcion	123 precio	123 impuesto
1	1.000	Varilla 3/16	100	10
2	1.010	Varilla 4/32	115	11,5
3	1.020	Varilla 3/17	130	13
4	1.030	Varilla 4/33	145	14,5
5	1.040	Varilla 3/18	160	16
6	1.050	Varilla 4/34	175	17,5
7	1.060	Varilla 3/19	190	19
8	1.070	Varilla 4/35	205	20,5
9	1.080	Ladrillos rojos	50	5
10	1.090	Ladrillos grises	35	3,5
11	1.100	Block	30	3
12	1.110	Megablock	40	4
13	1.120	Sillar rosa	100	10
14	1.130	Sillar gris	110	11
15	1.140	Cantera blanca	200	20
16	1.150	Cantera gris	1.210	121
17	1.160	Cantera rosa	1.420	142
18	1.170	Cantera amarilla	230	23
19	1.180	Recubrimiento P1001	200	20
20	1.190	Recubrimiento P1010	220	22
21	1.200	Recubrimiento P1019	240	24

Renovar Save Cancel Exportar datos ... 200 45 45 row(s) fetched

Comparación temporal

DBeaver	Supabase
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Selección

The screenshot shows the DBeaver interface. The top pane displays the SQL editor with the following query:

```
select * from "Materiales" m
where clave=1000
```

The bottom pane shows the data grid for the query. The columns are: clave, descripcion, precio, impuesto. The first row of data is:

clave	descripcion	precio	impuesto
1.000	Varilla 3/16	100	10

Comparación temporal

The image displays two side-by-side screenshots of application status windows. The left window, titled 'DBeaver', shows a message box with the text '1 row(s) fetched - 0,004s, on 2026-04-28 at 14:25:49'. The right window, titled 'Supabase', shows a similar message box with the text '1 row(s) fetched - 0,138s, on 2026-04-28 at 14:26:36'. Both windows feature a standard title bar with minimize, maximize, and close buttons, and a 'Aceptar' button at the bottom right.

Proyección

public *<postgres 2> Supabase × *<postgres> DBeaver

```
select clave,rfc,fecha from "Entregan" e
```

Entregan 1 ×

select * from "Materiales" m where clave=100 Enter a SQL expression to filter results (use Ctrl+Space)

	123 clave	AZ rfc	fecha
1	1.000	AAAA800101	1999-07-13
2	1.010	BBBB800101	1998-07-28
3	1.010	BBBB800101	1997-02-09
4	1.020	CCCC800101	2003-12-16
5	1.020	CCCC800101	2000-03-29
6	1.030	DDDD800101	1998-01-12
7	1.030	DDDD800101	2005-06-07
8	1.040	EEEE800101	1999-12-18
9	1.040	EEEE800101	1999-05-29
10	1.050	FFFF800101	2004-02-07
11	1.050	FFFF800101	2000-04-18
12	1.060	GGGG800101	2004-09-29
13	1.060	GGGG800101	2002-07-23
14	1.070	HHHH800101	2006-11-21
15	1.070	HHHH800101	2004-11-27
16	1.080	AAAA800101	2005-04-03
17	1.080	AAAA800101	2003-08-28
18	1.090	BBBB800101	1997-03-13
19	1.090	BBBB800101	1998-11-17

Renovar Save Cancel Exportar datos ... 200 86 86 row(s) fetched

Comparación temporal

The image displays two side-by-side screenshots of application status windows. The left window, titled 'DBeaver', shows a message indicating that 86 rows were fetched in 0.011 seconds (0.006 seconds for the fetch operation) on 2026-04-28 at 14:28:22. The right window, titled 'Supabase', shows a similar message but with a longer fetch time of 0.085 seconds (0.002 seconds for the fetch operation) on 2026-04-28 at 14:27:45. Both windows feature a 'Status details' title bar with standard window controls and an 'Aceptar' button at the bottom right.

Reunión Natural

*<postgres 2> Supabase

*<postgres> DBeaver X

select * from "Materiales" m , "Entregan" e
where m.clave = e.clave

Materiales(+) 1 X

select clave,rfc,fecha from "Entregan" e

Enter a SQL expression to filter results (use Ctrl+Space)

Grilla

Record

Texto

	123 clave	AZ descripcion	123 precio	123 impuesto	123 clave	AZ rfc	123 numero	123 fecha	123 cantidad
1	1.000	Varilla 3/16	100	10	1.000	AAAA800101	5.019	1999-07-13	
2	1.010	Varilla 4/32	115	11,5	1.010	BBBB800101	5.018	1997-02-09	
3	1.010	Varilla 4/32	115	11,5	1.010	BBBB800101	5.001	1998-07-28	
4	1.020	Varilla 3/17	130	13	1.020	CCCC800101	5.017	2000-03-29	
5	1.020	Varilla 3/17	130	13	1.020	CCCC800101	5.002	2003-12-16	
6	1.030	Varilla 4/33	145	14,5	1.030	DDDD800101	5.016	2005-06-07	
7	1.030	Varilla 4/33	145	14,5	1.030	DDDD800101	5.003	1998-01-12	
8	1.040	Varilla 3/18	160	16	1.040	EEEE800101	5.015	1999-05-29	
9	1.040	Varilla 3/18	160	16	1.040	EEEE800101	5.004	1999-12-18	
10	1.050	Varilla 4/34	175	17,5	1.050	FFFF800101	5.014	2000-04-18	
11	1.050	Varilla 4/34	175	17,5	1.050	FFFF800101	5.005	2004-02-07	
12	1.060	Varilla 3/19	190	19	1.060	GGGG800101	5.013	2002-07-23	
13	1.060	Varilla 3/19	190	19	1.060	GGGG800101	5.006	2004-09-29	
14	1.070	Varilla 4/35	205	20,5	1.070	HHHH800101	5.012	2004-11-27	
15	1.070	Varilla 4/35	205	20,5	1.070	HHHH800101	5.007	2006-11-21	
16	1.080	Ladrillos rojos	50	5	1.080	AAAA800101	5.011	2003-08-28	
17	1.080	Ladrillos rojos	50	5	1.080	AAAA800101	5.008	2005-04-03	

Renovar

Save

Cancel

Exportar datos ...

200

86

86 row(s) fetched - 0.013s (0.01s fetch), on 2026-04-28 at 14:29:29

Comparación temporal

The image displays two side-by-side screenshots of application status windows. The left window, titled 'DBeaver', shows a message indicating that 86 rows were fetched in 0.013 seconds on April 28, 2026, at 14:29:29. The right window, titled 'Supabase', shows a similar message but with a longer fetch time of 0.076 seconds on April 28, 2026, at 14:30:29. Both windows feature a title bar with standard window controls and a button labeled 'Aceptar' at the bottom right.

Reunión con criterio específico

public | * <postgres 2> Supabase X | * <postgres> DBaver

```
select * from "Entregan" e , "Proyectos" p
where e.numero <= p.numero
```

Entregan(+) 1 X

Enter a SQL expression to filter results (use Ctrl+Space)

	123 clave	AZ rfc	123 numero	fecha	123 cantidad	123 numero	AZ denominacion
1	1.000	AAAA800101	5.019	1999-07-13 GMT-C	254	5.019	Queretaro limpio
2	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.001	Aztecon
3	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.002	CIT Campeche
4	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.003	Mexico sin ti no estamos completos
5	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.004	Educando en Coahuila
6	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.005	Infonavit Durango
7	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.006	Reconstrucción del templo de Guadalupe
8	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.007	Construcción de plaza Magnolias
9	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.008	Televisa en acción
10	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.009	Disco Atlantic
11	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.010	Construcción de Hospital Infantil
12	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.011	Remodelación de aulas del IPP
13	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.012	Restauración de instalaciones del CEA
14	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.013	Reparación de la plaza Sonora
15	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.014	Remodelación de Soriana
16	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.015	CIT Yucatan
17	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.016	Ampliación de la carretera a la huasteca
18	1.010	B8B8800101	5.001	1998-07-28 GMT-C	528	5.017	Reparación de la carretera del sol

Renovar Save Cancel Exportar datos ... 200 200+

... 200 row(s) fetched - 0.137s (0.003s fetch), on 2026-04-28 at 14:32:41

Comparación temporal

Unión (se ilustra junto con selección)

The screenshot shows the DBeaver SQL editor with a query using the UNION operator. The query is:

```
(select * from "Entregan" e where clave=1450)
union
(select * from "Entregan" e where clave=1300)
```

Below the editor, the 'Resultados 1' tab displays the query results in a table format. The table has 7 columns: clave, rfc, numero, fecha, and cantidad. The results are as follows:

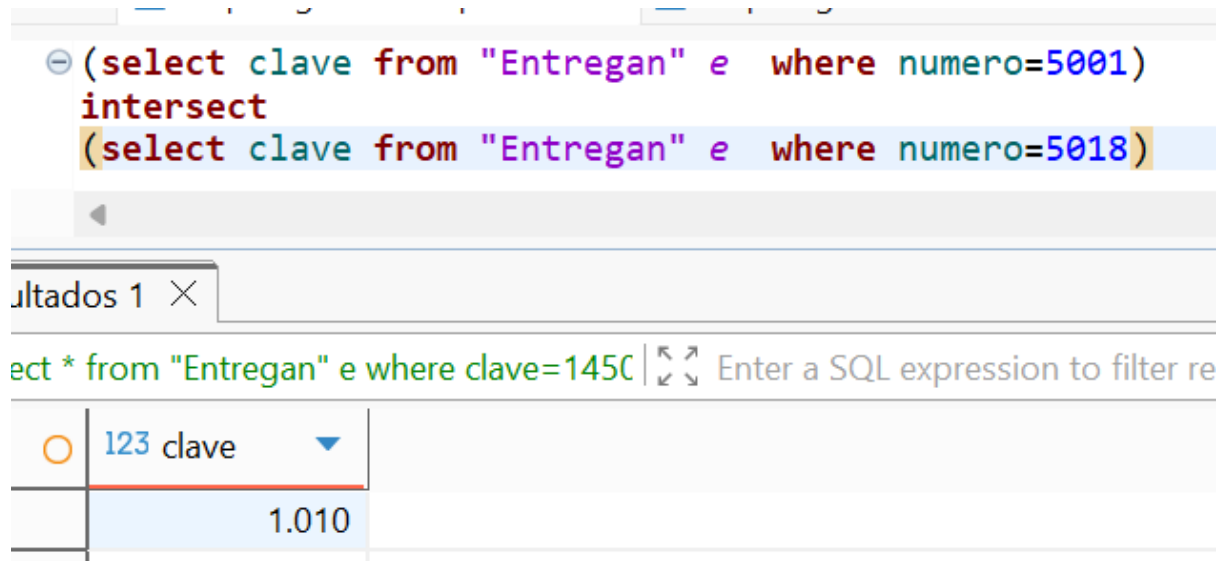
	clave	rfc	numero	fecha	cantidad
1	1.300	GGGG800101	5.005	2004-02-28	521
2	1.300	GGGG800101	5.010	2001-02-10	119

Comparación temporal

The image compares the status detail windows of DBeaver and Supabase. Both windows show the message: '2 row(s) fetched - 0,004s (0,001s fetch), on 2026-04-28 at 14:36:12' for DBeaver and '2 row(s) fetched - 0,087s (0,001s fetch), on 2026-04-28 at 14:35:56' for Supabase. Both windows have an 'Aceptar' button.

¿Cuál sería una consulta que obtuviera el mismo resultado sin usar el operador Unión?

Intersección (se ilustra junto con selección y proyección)



Comparación temporal

The image displays two side-by-side screenshots of application status windows. The left window, titled 'DBeaver', contains a message box stating '1 row(s) fetched - 0,003s, on 2026-04-28 at 14:38:40'. The right window, titled 'Supabase', contains a message box stating '1 row(s) fetched - 0,069s, on 2026-04-28 at 14:37:58'. Both windows feature a standard title bar with minimize, maximize, and close buttons, and an 'Aceptar' button at the bottom right.

Diferencia (se ilustra con selección)

DBeaver | <postgres> DBeaver | <postgres 2> Supabase X

```
(select * from "Entregan" e )
except
(select * from "Entregan" e | where clave = 1000);
```

Resultados 1 X

(select * from entregan) except (select * from ... Enter a SQL expression to filter results (use Ctrl+Space)

	123 clave	AZ rfc	123 numero	fecha	123 cantidad
1	1.340	CCCC800101	5.016	1997-05-11	674
2	1.270	DDDD800101	5.008	2003-01-21	546
3	1.430	DDDD800101	5.003	2005-04-30	576
4	1.200	EEEE800101	5.000	2003-03-15	177
5	1.300	GGGG800101	5.005	2004-02-28	521
6	1.350	DDDD800101	5.015	1997-01-06	272
7	1.200	EEEE800101	5.015	1998-11-07	585
8	1.070	HHHH800101	5.007	2006-11-21	2
9	1.120	EEEE800101	5.012	1997-08-05	215
10	1.370	FFFF800101	5.013	1997-12-03	575
11	1.410	BBBB800101	5.009	2003-07-13	467
12	1.010	BBBB800101	5.001	1998-07-28	528
13	1.140	GGGG800101	5.005	2000-06-30	583
14	1.020	CCCC800101	5.017	2000-03-29	8
15	1.140	GGGG800101	5.014	1999-06-28	219
16	1.370	FFFF800101	5.017	1999-06-26	44
17	1.400	AAAA800101	5.000	1999-04-07	382
18	1.310	HHHH800101	5.011	2000-09-14	72
19	1.030	DDDD800101	5.003	1998-01-12	202

Renovar Save Cancel Exportar datos ... 200 85 85 row(s) fetched

Comparación temporal

The image displays two side-by-side screenshots of application status windows. The left window, titled 'DBeaver', contains a message box stating '85 row(s) fetched - 0,018s (0,002s fetch), on 2026-05-03 at 12:41:17'. The right window, titled 'Supabase', contains a message box stating '85 row(s) fetched - 0,096s (0,004s fetch), on 2026-05-03 at 12:40:18'. Both windows feature a standard title bar with minimize, maximize, and close buttons, and an 'Aceptar' button at the bottom right.

Producto cartesiano

<postgres> DBeaver
 <postgres 2> Supabase

```
select * from "Entregan" e , "Materiales" m
```

Entregan(+) 1 X

(select * from "Entregan" e) except (select * f
 Enter a SQL expression to filter results (use Ctrl+Space)

	123 clave	AZ rfc	123 numero	fecha	123 cantidad	123 clave	AZ descripcion	123 precio	123 impuesto
1	1.000	AAAA800101	5.019	1999-07-13	254	1.000	Varilla 3/16	100	
2	1.010	BBBB800101	5.001	1998-07-28	528	1.000	Varilla 3/16	100	
3	1.010	BBBB800101	5.018	1997-02-09	523	1.000	Varilla 3/16	100	
4	1.020	CCCC800101	5.002	2003-12-16	582	1.000	Varilla 3/16	100	
5	1.020	CCCC800101	5.017	2000-03-29	8	1.000	Varilla 3/16	100	
6	1.030	DDDD800101	5.003	1998-01-12	202	1.000	Varilla 3/16	100	
7	1.030	DDDD800101	5.016	2005-06-07	295	1.000	Varilla 3/16	100	
8	1.040	EEEE800101	5.004	1999-12-18	263	1.000	Varilla 3/16	100	
9	1.040	EEEE800101	5.015	1999-05-29	540	1.000	Varilla 3/16	100	
10	1.050	FFFF800101	5.005	2004-02-07	503	1.000	Varilla 3/16	100	
11	1.050	FFFF800101	5.014	2000-04-18	623	1.000	Varilla 3/16	100	
12	1.060	GGGG800101	5.006	2004-09-29	324	1.000	Varilla 3/16	100	
13	1.060	GGGG800101	5.013	2002-07-23	692	1.000	Varilla 3/16	100	
14	1.070	HHHH800101	5.007	2006-11-21	2	1.000	Varilla 3/16	100	
15	1.070	HHHH800101	5.012	2004-11-27	503	1.000	Varilla 3/16	100	
16	1.080	AAAA800101	5.008	2005-04-03	86	1.000	Varilla 3/16	100	
17	1.080	AAAA800101	5.011	2003-08-28	699	1.000	Varilla 3/16	100	

Renovar
 Save
 Cancel

 Exportar datos ...
 200 200+

200 row(s) fetched - 0.026s (0.009s fetch), on 2026-05-03 at 12:42:31

Comparación temporal

The image displays two side-by-side screenshots of application status windows. The left window, titled 'DBeaver', shows a message indicating that 200 rows were fetched in 0.026 seconds (0.009 seconds for the fetch itself) on May 3, 2026, at 12:42:38. The right window, titled 'Supabase', shows a similar message but with a longer fetch time of 0.218 seconds (0.016 seconds for the fetch itself) on May 3, 2026, at 12:43:21. Both windows feature a standard title bar with a close button and a scrollable message area, with an 'Aceptar' button at the bottom right.

¿Cómo está definido el número de tuplas de este resultado en términos del número de tuplas de entregan y de materiales?

En el producto cartesiano cada tupla de entregan se combina con cada tupla de materiales.

Entonces, si: entregan tiene n tuplas y materiales tiene m tuplas el resultado tendrá:

 $n \times m$

Construcción de consultas a partir de una especificación

The screenshot shows the DBeaver application interface. At the top, there are two database connection tabs: "<postgres> DBeaver" and "<postgres 2> Supabase". The SQL editor displays the following query:

```
select distinct m.descripcion
from "Entregan" e , "Materiales" m
where e.clave = m.clave
and e.fecha >= '2000-01-01'
```

Below the SQL editor, a tab labeled "Materiales 1" is active. The results pane shows a table with the following data:

	AZ descripcion
1	Pintura C1010
2	Pintura B1022
3	Recubrimiento P1028
4	Arena
5	Varilla 3/17
6	Tubería 3.6
7	Pintura B1021
8	Sillar gris
9	Block
10	Varilla 4/34
11	Cantera blanca

On the left side of the results pane, there are two tabs: "Grilla" (Grid) and "Texto" (Text), with "Grilla" currently selected.

Comparación temporal

The image displays two side-by-side screenshots of application status windows. The left window, titled 'DBeaver', shows a message indicating that 11 rows were fetched in 0.002 seconds on May 3, 2026, at 12:49:14. The right window, titled 'Supabase', shows a message indicating that 11 rows were fetched in 0.11 seconds on May 3, 2026, at 12:48:56. Both windows feature a title bar with a status icon, the window title, and standard window controls (minimize, maximize, close). A scrollable text area contains the message, and an 'Aceptar' button is located at the bottom right of each window.

Ordenamientos

The screenshot shows the DBeaver application window with two tabs: '*<postgres> DBeaver' and '*<postgres 2> Supabase'. The active tab displays a SQL query:

```
select p.numero, p.denominacion, e.fecha, e.cantidad
from "Proyectos" p , "Entregan" e
where p.numero = e.numero
order by p.numero, e.fecha desc;
```

Below the query editor, the results are displayed in a table view. The table has columns: numero, denominacion, fecha, and cantidad. The results show 16 rows of data, ordered by numero and then by fecha descending.

numero	denominacion	fecha	cantidad
1	Vamos Mexico	2003-03-15	177
2	Vamos Mexico	1999-04-07	382
3	Aztecon	2000-05-21	43
4	Aztecon	2000-05-18	601
5	Aztecon	1998-07-28	528
6	CIT Campeche	2005-07-03	24
7	CIT Campeche	2003-12-16	582
8	CIT Campeche	2001-09-09	603
9	Mexico sin ti no estamos completos	2005-04-30	576
10	Mexico sin ti no estamos completos	1998-09-12	530
11	Mexico sin ti no estamos completos	1998-01-12	202
12	Educando en Coahuila	2002-11-14	453
13	Educando en Coahuila	2000-01-01	152
14	Educando en Coahuila	1999-12-18	263
15	Infonavit Durango	2004-02-28	521
16	Infonavit Durango	2004-02-07	503

At the bottom of the window, it indicates: '86 row(s) fetched - 0.047s (0.001s fetch), on 2026-05-0'.

Comparación temporal

The image shows two side-by-side screenshots of status message windows. The left window is titled 'DBeaver' and the right window is titled 'Supabase'. Both windows have a title bar with a close button, a maximize button, and a standard icon. The message in the DBeaver window reads: 'Message: 86 row(s) fetched - 0,047s (0,001s fetch), on 2026-05-03 at 12:52:31'. The message in the Supabase window reads: 'Message: 86 row(s) fetched - 0,261s (0,003s fetch), on 2026-05-03 at 12:53:54'. Both windows have a scrollable text area and a blue 'Aceptar' button at the bottom right.

Uso de expresiones.

*<postgres> DBeaver X *<postgres 2> Supabase X

```
-- select clave, cantidad, cantidad * 2 as doble_cantidad  
-- from "Entregan" e n;
```

Proyectos(+) 1 X

select p.numero, p.denominacion, e.fecha, e.cantidad

	123 numero	AZ denominacion	fecha	123 cantidad
1	5.000	Vamos Mexico	2003-03-15	177
2	5.000	Vamos Mexico	1999-04-07	382
3	5.001	Aztecon	2000-05-21	43
4	5.001	Aztecon	2000-05-18	601
5	5.001	Aztecon	1998-07-28	528
6	5.002	CIT Campeche	2005-07-03	24
7	5.002	CIT Campeche	2003-12-16	582
8	5.002	CIT Campeche	2001-09-09	603
9	5.003	Mexico sin ti no estamos completos	2005-04-30	576
10	5.003	Mexico sin ti no estamos completos	1998-09-12	530
11	5.003	Mexico sin ti no estamos completos	1998-01-12	202
12	5.004	Educando en Coahuila	2002-11-14	453
13	5.004	Educando en Coahuila	2000-01-01	152
14	5.004	Educando en Coahuila	1999-12-18	263
15	5.005	Infonavit Durango	2004-02-28	521
16	5.005	Infonavit Durango	2004-02-07	503
17	5.005	Infonavit Durango	2003-10-12	71
18	5.005	Infonavit Durango	2000-06-30	583

Renovar Save Cancel Exportar datos ... 200

86 row(s) fetched

Comparación temporal

Operadores de cadena

The screenshot shows two database interfaces side-by-side. The left interface is DBeaver, with a SQL query editor containing: `SELECT * FROM "Materiales" m where m.descripcion LIKE 'Si%'`. Below the editor, a table titled 'Materiales 1' displays the results of the query. The right interface is Supabase, showing a SQL query editor with a partial query: `SELECT * FROM "Proyectos" p where Descrip`.

	123 clave	AZ descripcion	123 precio	123 impuesto
1	1.120	Sillar rosa	100	10
2	1.130	Sillar gris	110	11

Comparación temporal

DBeaver	Supabase

¿Qué resultado obtienes?

Explica que hace el símbolo '%'.
¿Qué sucede si la consulta fuera : LIKE 'Si' ?

¿Qué resultado obtienes?

Explica a qué se debe este comportamiento.

La consulta LIKE 'Si%' obtiene todos los registros cuya descripción empieza con "Si", porque el símbolo % funciona como comodín y representa cualquier cadena de cero o más caracteres después de "Si"; por ejemplo, podría encontrar "Silla", "Sillón" o "Silicón". En cambio, si la consulta fuera LIKE 'Si', solo devolvería registros cuya descripción sea exactamente "Si", ya que al no incluir %, SQL no permite que haya más caracteres antes o después de esa cadena; por eso normalmente no regresaría resultados, a menos que exista un producto llamado literalmente "Si".

Modificando la estructura de una tabla existente.

*<postgres> DBeaver

*<postgres 2> Supabase

```
select e.clave,
       m.descripcion,
       e.cantidad,
       m.precio,
       m.PorcentajeImpuesto,
       e.cantidad * m.precio * (1 + m.PorcentajeImpuesto / 100) as importe
from "Entregan" e, "Materiales" m
where e.clave = m.clave;
```

Entregan(+)

select * from "Materiales" m

	123 clave	AZ descripcion	123 cantidad	123 precio	123 porcentajeimpuesto	123 importe
1	1.000	Varilla 3/16	254	100	2	25.908
2	1.010	Varilla 4/32	523	115	2	61.347,9
3	1.010	Varilla 4/32	528	115	2	61.934,4
4	1.020	Varilla 3/17	8	130	2	1.060,8
5	1.020	Varilla 3/17	582	130	2	77.173,2
6	1.030	Varilla 4/33	295	145	2	43.630,5
7	1.030	Varilla 4/33	202	145	2	29.875,8
8	1.040	Varilla 3/18	540	160	2	88.128
9	1.040	Varilla 3/18	263	160	2	42.921,6
10	1.050	Varilla 4/34	623	175	2	111.205,5
11	1.050	Varilla 4/34	503	175	2	89.785,5
12	1.060	Varilla 3/19	692	190	2	134.109,6
13	1.060	Varilla 3/19	324	190	2	62.791,2
14	1.070	Varilla 4/35	503	205	2	105.177,3
15	1.070	Varilla 4/35	2	205	2	418,2

Renovar Save Cancel Exportar datos ... 200 86

86 row(s) fetched - 0,007s (0,003s fetch), on 2026-05-03 at 13:22:32

Comparación temporal

DBeaver	Supabase
Status details Message: 86 row(s) fetched - 0,007s (0,003s fetch), on 2026-05-03 at 13:22:32 Aceptar	Status details Message: 86 row(s) fetched - 0,197s, on 2026-05-03 at 13:23:24 Aceptar